DevOps Capstone Projects

Beginner Edition: Ship It and Keep It Running

Presented by
Toluwalase Adedotun

Two Projects, Step by Step

Capstone 1: Ship It

- Modules 5 to 12
- Docker, Terraform, GitHub Actions, Helm
- Goal: from code to automatic deployment

Capstone 2: Keep It Running

- Modules 13 and 14
- Prometheus, Grafana, Fluent Bit, OpenTelemetry
- Goal: a service you can watch and trust

How It Works

- Individual work, not groups
- Fork your own repo
- Catch-ups every Wednesday and Saturday
- Final demo and Q&A

Capstone 1: The Voting App

Services

- vote: the page where you cast a vote
- result: the page showing live totals
- worker: saves votes to the database
- Redis (queue) and PostgreSQL (database)

Data Flow

- You vote in the vote app
- The vote waits in Redis
- The worker saves it to PostgreSQL
- The result app shows the totals

Your Starting Point

- Fork dockersamples/example-voting-app
- Run it with Docker Compose
- Work only in your own fork

Capstone 1: Five Steps

1. Run it locally

Fork the repo and run docker compose up. You can cast a vote and see it counted.

2. Containerize it safely

Pin base images, run as non-root, add a .dockerignore. Every image builds and starts.

3. Create a k8s cluster with Terraform

Terraform builds a Kubernetes (k8s) cluster in the default VPC. Destroy it, then recreate it from code.

4. Automate with GitHub Actions

Each push runs tests, a Trivy scan, a build and a push, tagged with the commit SHA.

5. Deploy and roll back with Helm

One chart, one values file. Ship a bad release on purpose, then run helm rollback.

Optional stretch

Ingress, remote Terraform state, Argo CD or autoscaling. Only if you finish early.

Capstone 1: Done Means

Hand In

- Forked repo with a clear README
- Terraform code and one Helm chart
- A working GitHub Actions pipeline
- Architecture diagram

Done Means

- One push builds, scans and deploys
- The environment rebuilds from code
- No secrets in Git
- Rollback is shown live

Capstone 2: Stan's Robot Shop

The Shop

- About a dozen small services
- Install it with the provided Helm chart
- Remove the Instana agent settings
- Use load-gen for steady traffic

Focus on Four

- web, catalogue, cart and payment
- The other services only need to run
- Data stores: MongoDB, Redis, MySQL, RabbitMQ

Where It Runs

- Your Capstone 1 cluster
- Fork instana/robot-shop first
- Work only in your own fork

Capstone 2: Six Steps

1. Deploy and watch

Install Robot Shop on your Capstone 1 cluster and start load-gen. All pods are healthy.

2. See the metrics

Install kube-prometheus-stack with Helm and build one dashboard for requests, errors and latency.

3. Collect the logs

Run Fluent Bit and send logs to Loki. You can search by service and error level.

4. Follow one trace

Use the instructor's OpenTelemetry config on one service and find one trace in Jaeger.

5. Set one SLO

Pick availability, write a one page error budget policy, and send one alert to Slack.

6. Run the incident drill

The instructor breaks something. Mitigate first, find the root cause, then write a postmortem.

Capstone 2: Done Means

Hand In

- Dashboard exported as code
- Logging and tracing config
- SLO document with error budget
- Blameless postmortem

Done Means

- The alert fires during the drill
- One request is traced
- Logs are searchable by service and level
- A runbook is linked from the alert

Optional

- Burn rate alerts
- Trace to log links
- Litmus chaos test

What the Instructor Gives You

Starter Kits

- README skeleton
- Terraform starter for a k8s cluster
- Helm chart skeleton

Ready Configs

- kube-prometheus-stack values
- Fluent Bit config for Loki
- OpenTelemetry Collector config

Templates

- Architecture diagram
- SLO and error budget policy
- Runbook
- Blameless postmortem

Words to Know

Word	Plain meaning
CI/CD	Build, test and deploy automatically on every push
Terraform	Code that creates your servers for you
Helm	Packages a Kubernetes app so you can install and roll back
SLI	A number that measures health, such as success rate
SLO	The target for that number, such as 99% success
Error budget	How much failure the SLO allows

Timeline and Grading

Capstone 1: 1 Week

- Days 1 and 2: steps 1 to 3
- Days 3 and 4: steps 4 and 5
- Day 5: demo

Capstone 2: 1 Week

- Days 1 and 2: steps 1 to 3
- Days 3 and 4: steps 4 and 5
- Day 5: drill and demo

Grading

- Working system: 50%
- Automation and docs: 30%
- Demo and Q&A: 20%

Submit your work

Open the submission form